



University
of Windsor

Comp-3150 (60-315) Winter 2020

Midterm 2 (For Practice 2026)

Examiner: Dr. C. I. Ezeife

Given: Thursday, March 5, 2020

Student Name: _____

Student Number: _____

INSTRUCTIONS (Please Read Carefully)

Examination Period is 1 hours 20 minutes

Answer all questions. Write your answers in the spaces provided in the question paper. This is closed book and closed notes test.

Total Marks =50. Total number of sections = 2

Please read questions carefully! Misinterpreting a question intentionally or unintentionally results in getting a “ZERO” for that question. Good Luck!!!

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CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY

I confirm that I will keep the content of this examination confidential.

I confirm that I have not received any unauthorized assistance in preparing for or writing this examination. I confirm knowing that a mark of 0 may be assigned for copied WORK.

Student Signature

Student Name (please print)

Student I.D. Number

Date

For marking purposes only (This part not to be filled by students)

Question	Mark
Section A (10 marks for 10 multiple choice questions)	/10
Section B	
Que 1 (10 marks)	/10
Que 2 (15 marks)	/15
Que 3 (15 marks)	/15
Total	/50

Section A

10 marks for 10 Multiple Choice Questions. Each question in this section is worth 1 mark.

Marking scheme: 1 mark for each correct multiple choice.

For Questions 1 – 10, circle the most correct answer for each question.

Consider a relation $R(\underline{V}, \underline{W}, X, Y, Z)$ where X is a candidate key and the following functional dependencies exist:

$(V, W) \rightarrow \{X, Y, Z\}$

$X \rightarrow \{V, W, Y, Z\}$

1. The relation $R(\underline{V}, \underline{W}, X, Y, Z)$ above is in 2NF:

- a. TRUE
- b. FALSE
- c. MAY BE
- d. ALL OF THE ABOVE
- e. NONE OF THE ABOVE

2. The relation $R(\underline{V}, \underline{W}, X, Y, Z)$ above is in 3NF:

- a. TRUE
- b. FALSE
- c. MAY BE
- d. ALL OF THE ABOVE
- e. NONE OF THE ABOVE

Given a relation $R1(\underline{V}, \underline{W}, \underline{X}, Y)$ with the following functional dependencies:

$(V, W, X) \rightarrow \{Y\}$

$Y \rightarrow X$

3. Relation $R1(\underline{V}, \underline{W}, \underline{X}, Y)$ is in BCNF:

- a. TRUE
- b. FALSE
- c. MAY BE
- d. ALL OF THE ABOVE
- e. NONE OF THE ABOVE

4. The relation $R1(\underline{V}, \underline{W}, \underline{X}, Y)$ above is in 3NF:

- a. TRUE
- b. FALSE
- c. MAY BE
- d. ALL OF THE ABOVE
- e. NONE OF THE ABOVE

Given the relation **Family(Mom, Dad, Pete, Paul)** with the following functional dependencies:

(Mom, Dad) -> {Pete, Paul}

Mom -> {Pete}

5. The relation **Family(Mom, Dad, Pete, Paul)** above is in 3NF:

- a. TRUE
- b. FALSE
- c. MAY BE
- d. ALL OF THE ABOVE
- e. NONE OF THE ABOVE

6. The relation **Family(Mom, Dad, Pete, Paul)** above is in 2NF:

- a. TRUE
- b. FALSE
- c. MAY BE
- d. ALL OF THE ABOVE
- e. NONE OF THE ABOVE

Given the following relation table state for PROJECT, where Pnumber is the primary key and Dnum refers to Dnum in another relation DEPARTMENT, use it to answer the next few applicable questions.

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	10	Bellaire	6
ProductY	11	Sugarland	6
ProductZ	12	Houston	6
Computerize	20	Stafford	3
Reorganize	40	Houston	1
Newbenefits	50	Stafford	3

7. Which of the following statements is correct about the table PROJECT?

- a. It contains update anomaly
- b. It contains insert anomaly
- c. It contains delete anomaly
- d. All of the above
- e. None of the above

8. The effect of the following SQL instruction on the PROJECT table, when the only known Dnum instances in the DEPARTMENT table are 1, 3 and 6 is _____ .

INSERT INTO PROJECT VALUES ('Finance', 30, 'Windsor', 2);

- a. Insert an additional tuple in the PROJECT table with values specified.
 - b. Update the second tuple of the PROJECT table to have the values specified.
 - c. Raise an exception to indicate that there is a referential integrity constraint violation.
 - d. Raise an exception to indicate that there is a primary key constraint violation.
 - e. None of the above
9. A basic SQL retrieval query to get all PROJECTs in Dnum 6 is:
- a. **Select * where Plocation = 'Sugarland';**
 - b. **Select * from PROJECT P where P.Dnum = 6;**
 - c. **Select * from PROJECT where Plocation ='Bellaire' or 'Sugarland' or 'Houston';**
 - d. All of the above
 - e. None of the above
10. To retrieve the number of PROJECTs handled by each Dnum in the table PROJECT, we can use the query:
- a. **Select Dnum, count(*) from PROJECT group by Dnum;**
 - b. **Select min(*) from PROJECT group by Dnum;**
 - c. **Select Dnum, sum(*) from PROJECT group by Dnum;**
 - d. **Select max(*) from PROJECT group by Dnum;**
 - e. None of the above

Section B (40 marks):**This section has 3 questions. Answer all questions in this section.**

1. **(10 marks)** Given the following relation schema T, with attribute A as the primary key, answer the questions below.

T Relation

<u>A</u>	B	C	D
a1	b1	c1	d1
a2	b2	c2	d1
a3	b2	c1	d1
a4	b2	c3	d1

- (i) What functional dependencies FD can you see exist in this relation above?
(ii) What update, (iii) delete and (iv) insertion anomalies may or not be present in this T relation above. Explain with specific examples using FDs and keys in this database.

Que 1 **(2.5 marks for each of i to iv for a total of 10 marks)**

(i) FDs (2.5 marks)	
(ii) Update anomalies (2.5 marks)	

(iii) Delete anomalies (2.5 marks)	
(iv) Insertion anomalies (2.5 marks)	

2. **(15 marks)** Given the following four relations for a simple hospital management record database application where prescriptions given to patients by doctors are tracked for each treatment. Total number of prescriptions received by each patient is known from the patient record at all times.

Patient(pid, pname, address, numprescr)
 Doctor(did, dname, specialization, address)
 Prescription(prescriid, prescr_name, qty)
 Treats(did, pid, prescriid, Pdate)

Here, the primary key for each relation is underlined. The attributes pid stands for patient id, did is doctor id, prescriid is prescription id, numprescr is the cumulative number of prescriptions the patient has received from doctors (e.g., 50). All other attributes are self explanatory.
 A small database state of this database is:

PATIENT

<u>pid</u>	pname	address	numprescr
1111	John Smith	Windsor	100
2222	Mary Pert	Windsor	50

DOCTOR

<u>did</u>	dname	specialization	address
1	Mana Patel	Family Physician	Windsor
2	Pat Goodman	Surgeon	London

PRESCRIPTION

<u>prescrip</u>	Prescr_name	Qty
23567	Inflam FX	2
55555	Tylenol	10

TREATS

<u>did</u>	<u>pid</u>	<u>prescrip</u>	Pdate
1	1111	23567	10-Jan-19
2	2222	55555	20-Aug-19

- Write the SQL DDL instructions to create all 4 tables of the above database in the Oracle DBMS on our cs server specifying all constraints such as Key, entity and foreign key constraints as usual within each instruction (not separately). **(5 marks)**
- Using only the data in the above instance, write the SQL instructions to load(insert) all the data above in the 4 tables you created. **(5 marks)**
- Write the SQL query for the following query posed on this database, also write the results of the query and the tables involved in answering the query. **(5 marks)**

Print all patients (pid, pname) treated by doctor 'Mana Patel'

(5 marks)

Que 2a **(5 marks)**: Creating Tables

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Que 2b (5 marks): Inserting Data into the Database Tables
Solution (b)

3. (15 marks)

Given the following WorkPlace relation schema, with Ssn as the primary key, answer the questions below.

WorkPlace

<u>Ssn</u>	name	locker#	rank	level	wage	hours
1	Bob	48	7	M	20	40
2	Bill	22	7	M	20	30
3	Pat	35	5	E	10	30
4	Tony	35	5	E	10	32
5	Namdi	35	7	M	20	40

- Is this table in 3NF?, If not in 3NF, normalize it into 3NF showing all the tables in your new decomposed database. Explain using functional dependencies. **(6 marks)**
- Provide one valid tuple instance that can be inserted into this original database. **(3 marks)**
- Provide one SQL query for updating a record in the original table. **(3 marks)**
- Provide one SQL query for (i) deleting all tuples in the original table, then (ii) provide one SQL query for deleting the original table schema from your database. **(3 marks)**

Question	Answers
a. (6 marks) Is this table in 3NF?, normalize into 3NF showing all tables Explain with FDs	
b. (3 marks)	

